



SYMBIOSIS INSTITUTE OF TECHNOLOGY

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NAAC with "A" Grade

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Batch – B3

COURSE – JAVA

Experiment 10-

Write a Java program to demonstrate the use of abstract classes and abstract methods

AIM – Write a Java program to demonstrate the use of abstract classes and abstract methods

Program 10

```
// Abstract class
```

```
abstract class Shape {

    // Abstract method
    abstract void draw();

    // Normal method
    void display() {
        System.out.println("This is a Shape.");
    }
}

// Child class
class Circle extends Shape {

    // Implementing the abstract method
    void draw() {
        System.out.println("Drawing a Circle.");
    }
}

// Child class
class Rectangle extends Shape {

    // Implementing the abstract method
    void draw() {
        System.out.println("Drawing a Rectangle.");
    }
}

// Main class
public class Main {

    public static void main(String[] args) {

        // Creating objects of child classes
        Circle c = new Circle();
        Rectangle r = new Rectangle();

        // Calling methods
        c.display();
        c.draw();

        r.display();
        r.draw();
    }
}
```

```
}
```

Output

```
PS C:\Users\cl404_17\javav\java> javac Main.java
PS C:\Users\cl404_17\javav\java> java Main.java
This is a Shape.
Drawing a Circle.
This is a Shape.
Drawing a Rectangle.
PS C:\Users\cl404_17\javav\java> |
```

ENG 11:58
IN 30-07-2026

EXERCISE

10.1 . .Create an abstract Payment class and implement Credit Card and UPI payment classes.?

Program 10.1 -

```
// Abstract class
abstract class Payment {
```

```
// Abstract method
abstract void pay();

// Normal method
void display() {
    System.out.println("Payment processing...");
}
}

// Child class
class CreditCard extends Payment {

    // Implementing abstract method
    void pay() {
        System.out.println("Payment made using Credit Card.");
    }
}

// Child class
class UPI extends Payment {

    // Implementing abstract method
    void pay() {
        System.out.println("Payment made using UPI.");
    }
}

// Main class
public class Main {

    public static void main(String[] args) {

        // Creating objects of child classes
        CreditCard c = new CreditCard();
        UPI u = new UPI();

        // Calling methods
        c.display();
        c.pay();

        u.display();
        u.pay();
    }
}
```

Screenshot 10.1 –

```
PS C:\Users\cl404_17\javav\java> javac Main.java
PS C:\Users\cl404_17\javav\java> java Main.java
Payment processing...
Payment made using Credit Card.
Payment processing...
Payment made using UPI.
PS C:\Users\cl404_17\javav\java> |
```

Program 10.2- Create an abstract FoodOrder class with an abstract method calculateBill(). Implement two subclasses, DineInOrder and TakeAwayOrder, to calculate and display the total bill based on the order type.

```
// Abstract class
abstract class FoodOrder {

    // Abstract method
    abstract void calculateBill();

    // Normal method
    void display() {
        System.out.println("Food Order Details");
    }
}
```

```
// Child class
class DineInOrder extends FoodOrder {

    // Implementing abstract method
    void calculateBill() {
        double foodPrice = 500;
        double serviceCharge = 50;

        double total = foodPrice + serviceCharge;

        System.out.println("Dine-In Bill: Rs. " + total);
    }
}

// Child class
class TakeAwayOrder extends FoodOrder {

    // Implementing abstract method
    void calculateBill() {
        double foodPrice = 500;
        double packingCharge = 30;

        double total = foodPrice + packingCharge;

        System.out.println("Take-Away Bill: Rs. " + total);
    }
}

// Main class
public class Main {

    public static void main(String[] args) {

        // Creating objects of child classes
        DineInOrder dineIn = new DineInOrder();
        TakeAwayOrder takeAway = new TakeAwayOrder();

        // Calling methods
        dineIn.display();
        dineIn.calculateBill();

        takeAway.display();
        takeAway.calculateBill();
    }
}
```

Screenshot 10.2 –

```
PS C:\Users\cl404_17\javav\java> javac Main.java
PS C:\Users\cl404_17\javav\java> java Main.java
Food Order Details
Dine-In Bill: Rs. 550.0
Food Order Details
Take-Away Bill: Rs. 530.0
PS C:\Users\cl404_17\javav\java> |
```